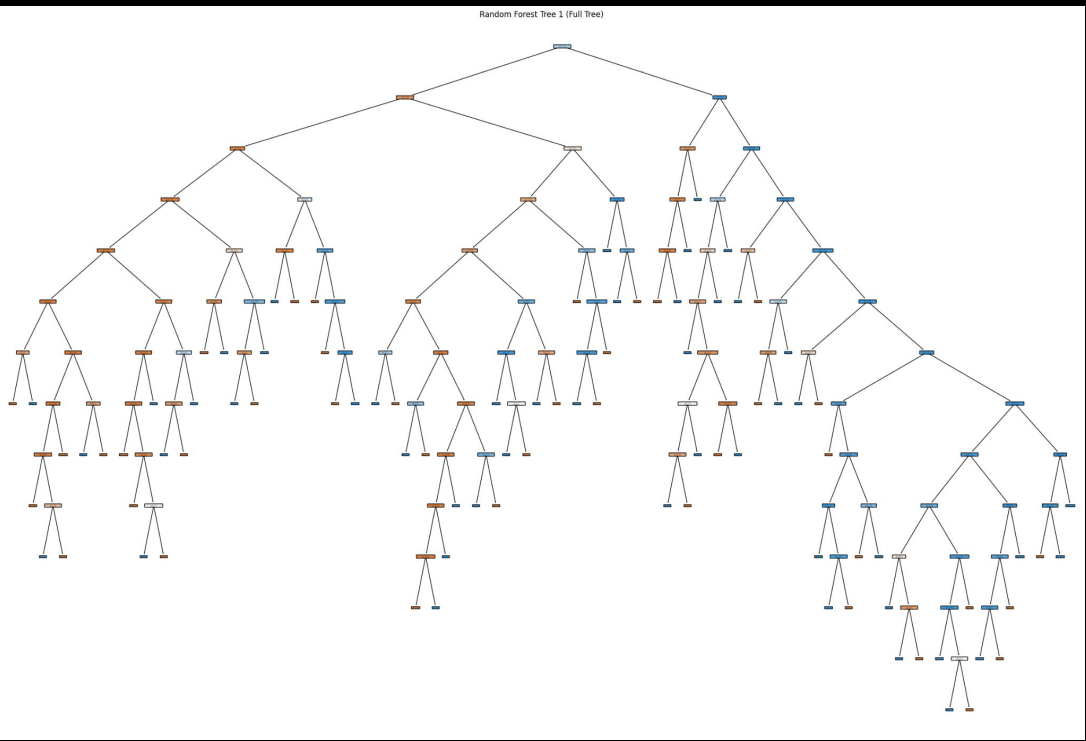


Random Forest Model - All Stations (2010s)

Question:
Which station's weather observations had the strongest influence on overall *pleasantness* in the 2010s?

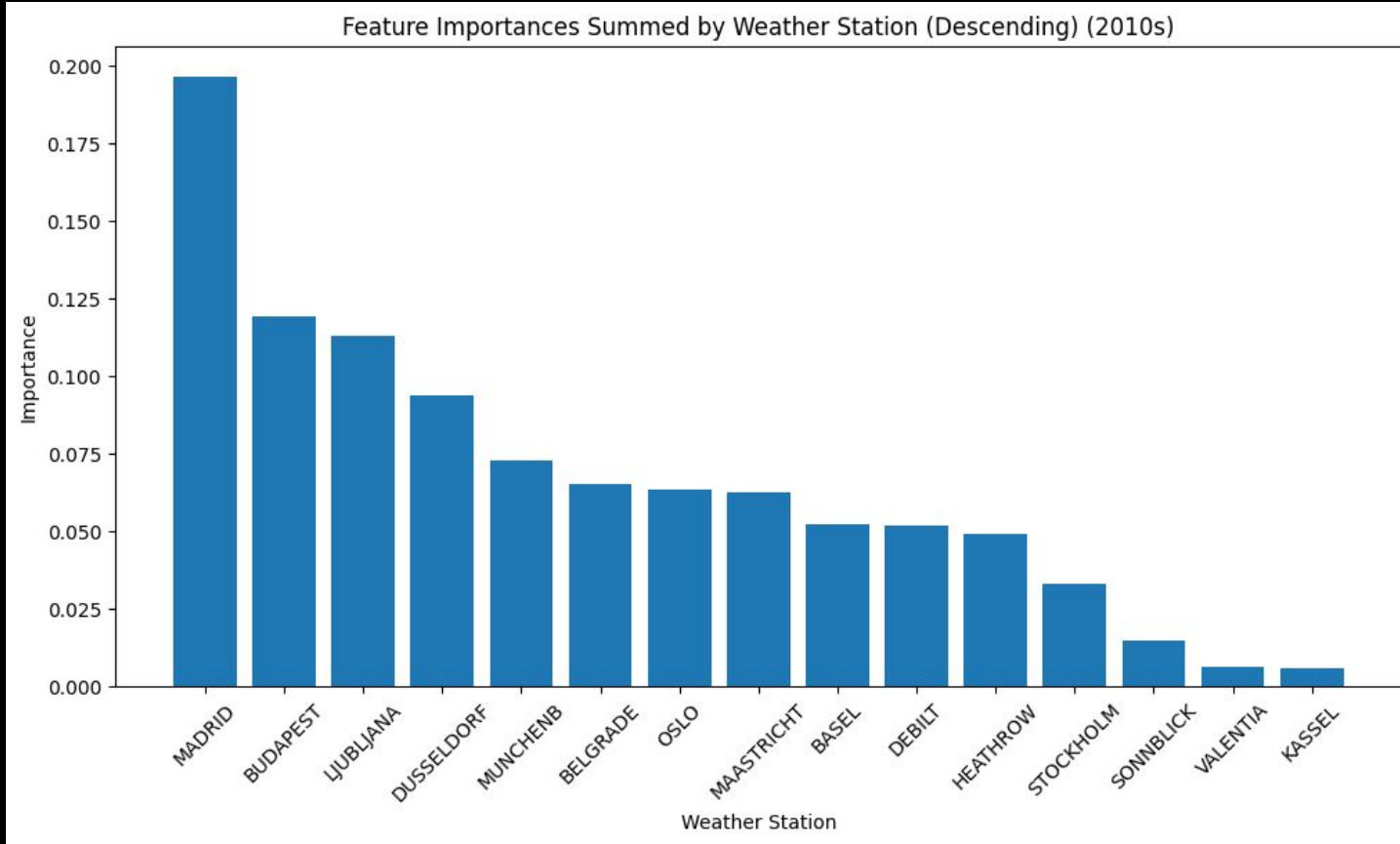
Method:
Trained a Random Forest classifier (`n_estimators = 100`; default scikit hyperparameter values) to predict pleasant vs. unpleasant days based on **9 weather features across 15 stations**.

Result:
Achieved **~97% accuracy**, despite the ambiguity of defining "pleasant" across varied locations and weather patterns.

Takeaway:
The model captures a consistent pattern of pleasantness, learned across diverse stations and weather dynamics.

Random Forest accuracy (all stations): 0.967

Random Forest Model - All Stations (2010s): Weather Stations Ranked by Influence on RF's Division of Data



Top 3:

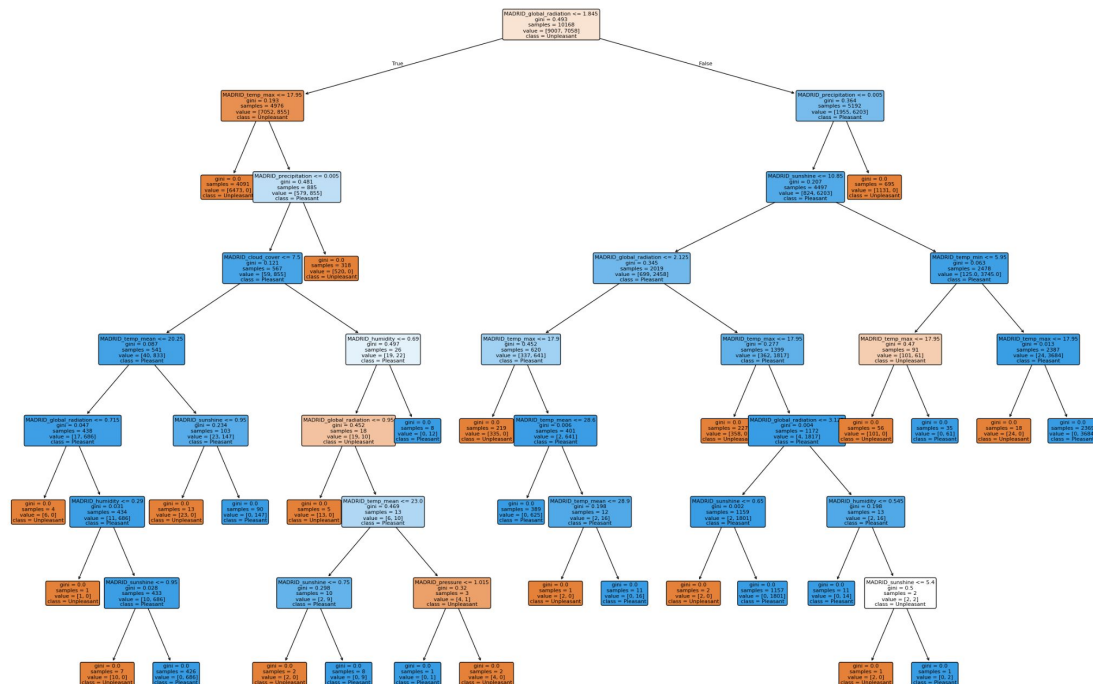
1) Madrid

2) Budapest

3) Ljubljana

Station Spotlight – MADRID

Full Decision Tree from Random Forest for MAD Pleasantness



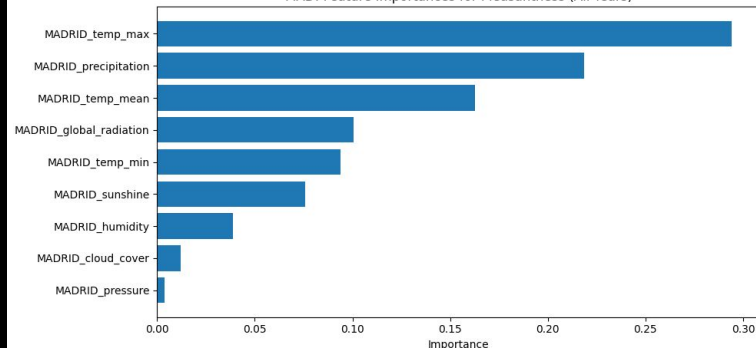
Model: Random Forest (Madrid, All Years; 100 trees & scikit defaults)

Accuracy: 100%

Top 3 predictors of “pleasant” days:

- temp_max
- precipitation
- temp_mean

MAD: Feature Importances for Pleasantness (All Years)

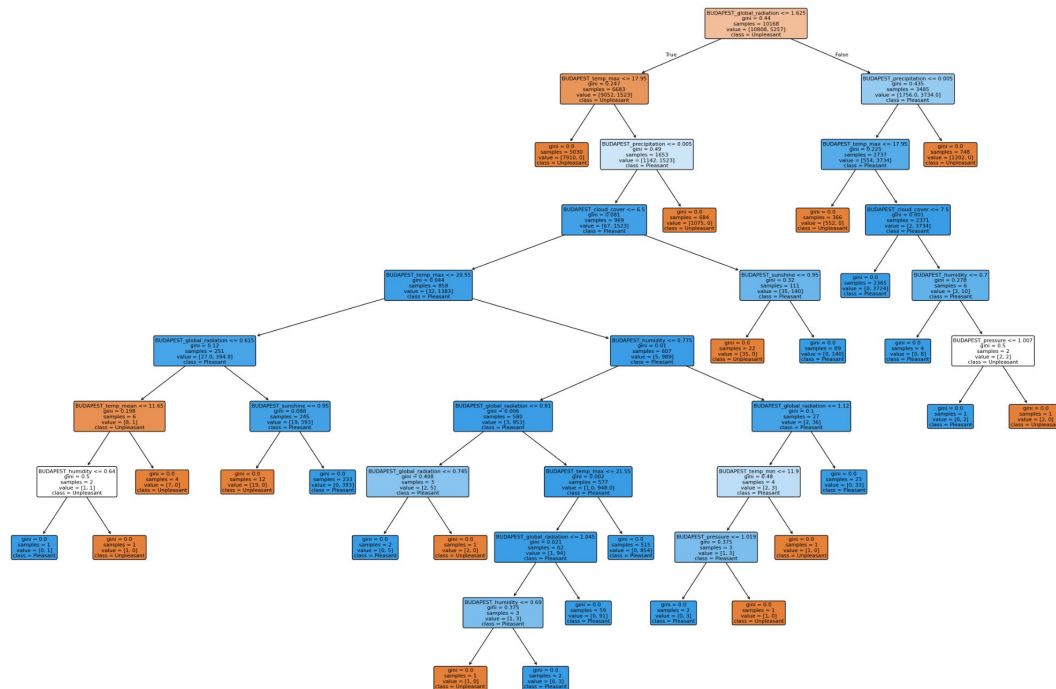


Training Random Forest for MADRID...



Accuracy for MADRID: 1.000

Station Spotlight – BUDAPEST



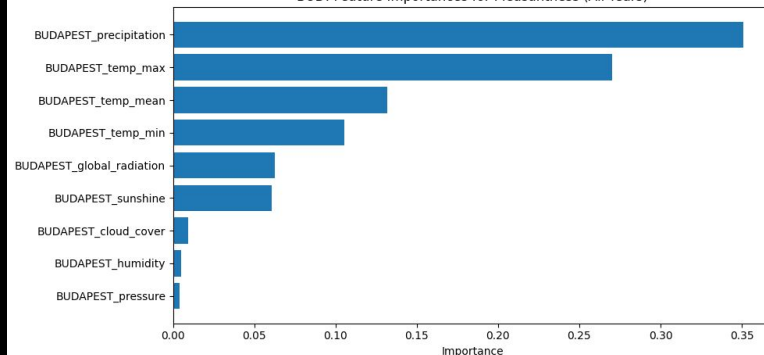
Model: Random Forest (Budapest, All Years; 100 trees & scikit defaults)

Accuracy: 100%

Top 3 predictors of “pleasant” days:

- precipitation
- temp_max
- temp_mean

BUD: Feature Importances for Pleasantness (All Years)



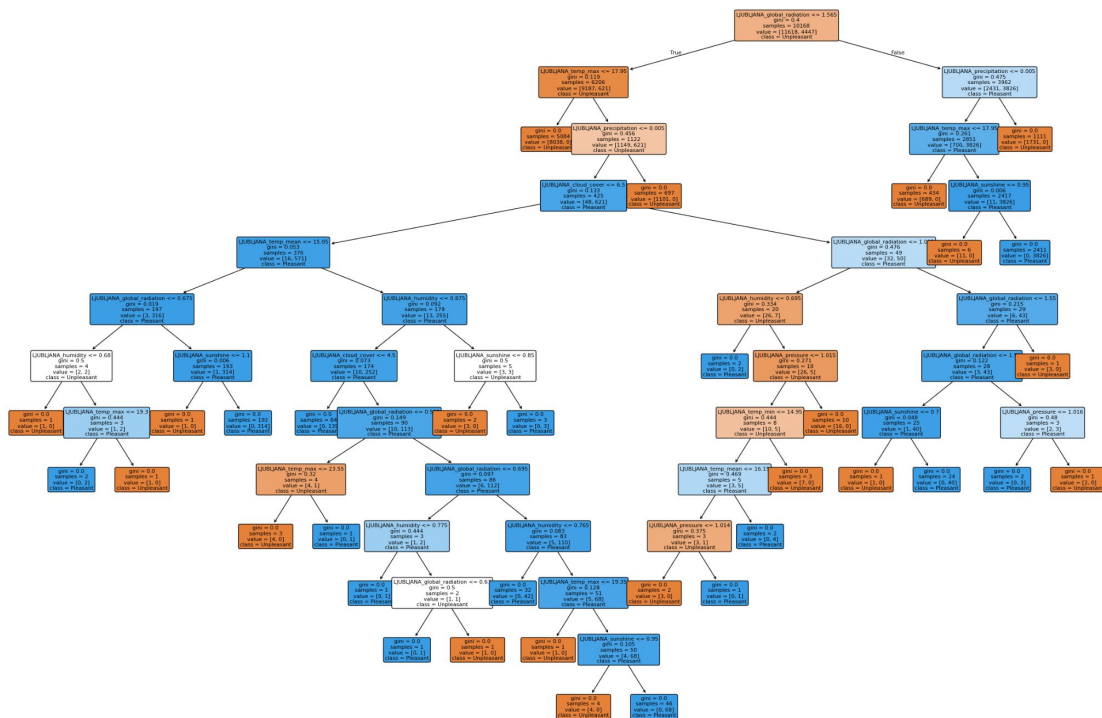

Training Random Forest for BUDAPEST...



Accuracy for BUDAPEST: 1.000

Station Spotlight – LJUBLJANA

Full Decision Tree from Random Forest for LJU Pleasantness



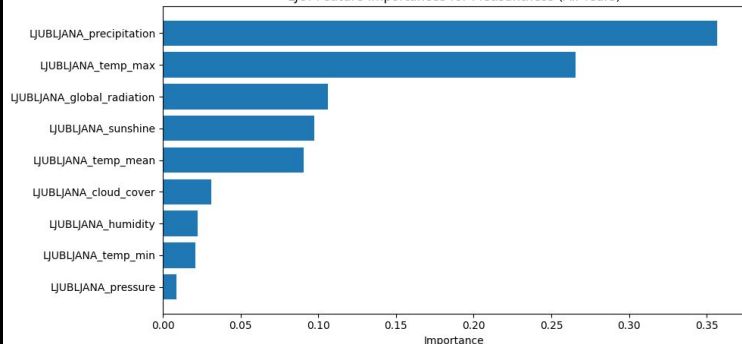
Model: Random Forest (Ljubljana, All Years; 100 trees & scikit defaults)

Accuracy: 100%

Top 3 predictors of “pleasant” days:

- precipitation
- temp_max
- global_radiation

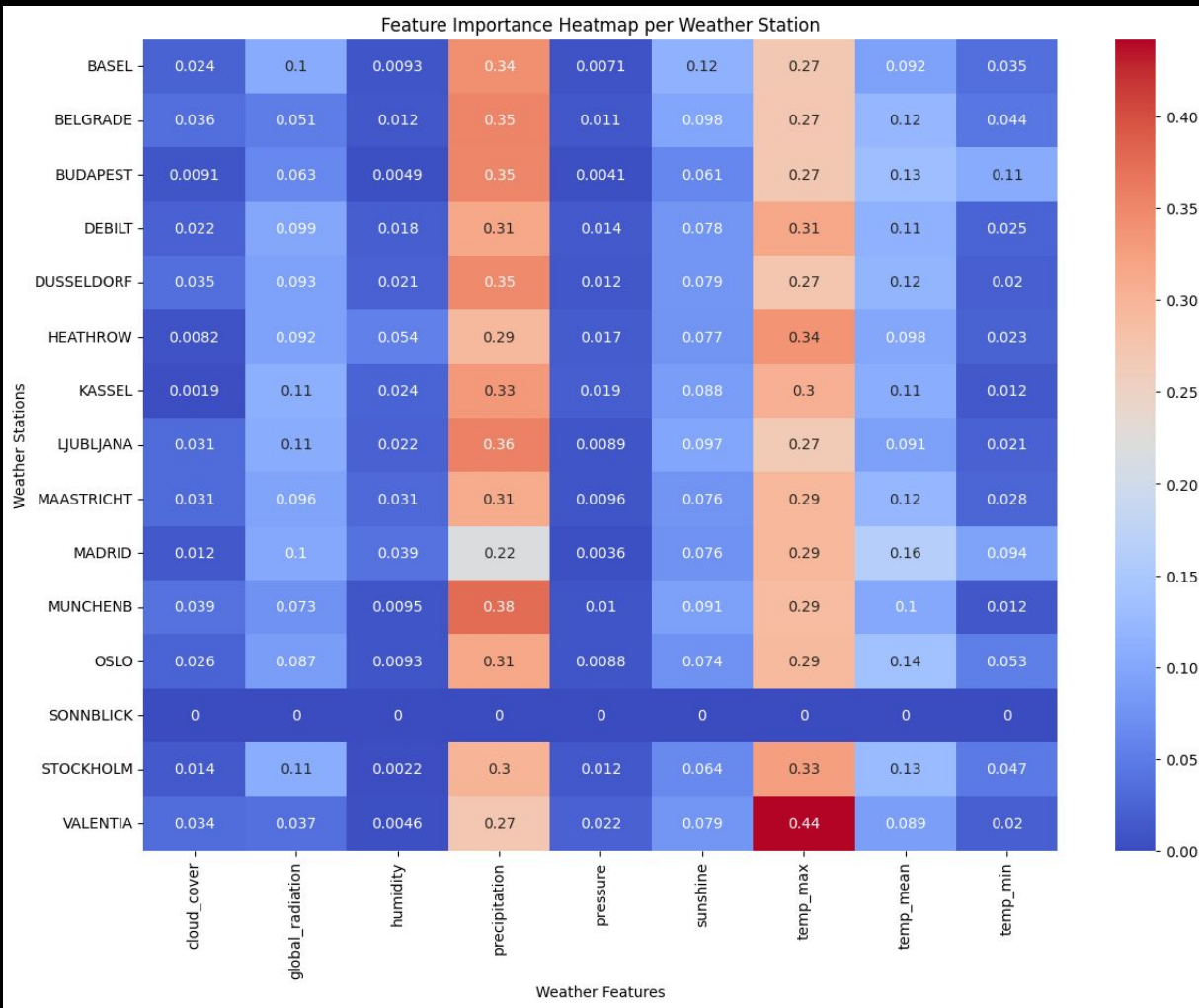
LJU: Feature Importances for Pleasantness (All Years)



Training Random Forest for LJUBLJANA...



Accuracy for LJUBLJANA: 1.000



Weather Fingerprints

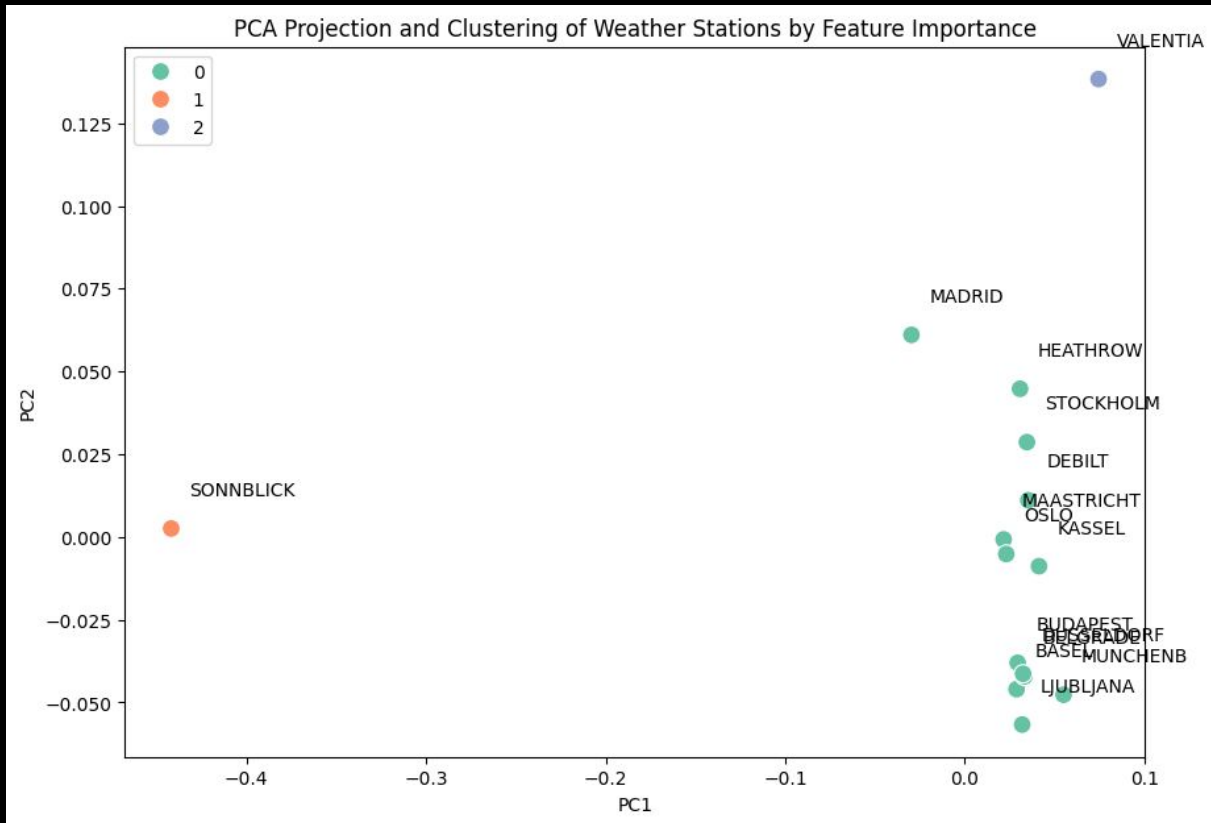
Each weather station's **ranking of weather metric importance** (as learned by its individual Random Forest model) can be treated as a “**weather fingerprint**.”

While most stations share a similar signature — with **precipitation** and **temp_max** consistently among the top drivers of pleasantness — **regional differences still emerge**:

- **Madrid** places significantly less importance on precipitation.
- **Valencia** ranks **temp_max** much higher than other stations.
- Smaller shifts in **sunshine**, **global_radiation**, and **cloud_cover** also show localized influence.

These “fingerprints” served as the basis for further dimensionality reduction and clustering in the next step: **PCA + Agglomerative Clustering**, to explore groups of stations with shared pleasantness patterns.

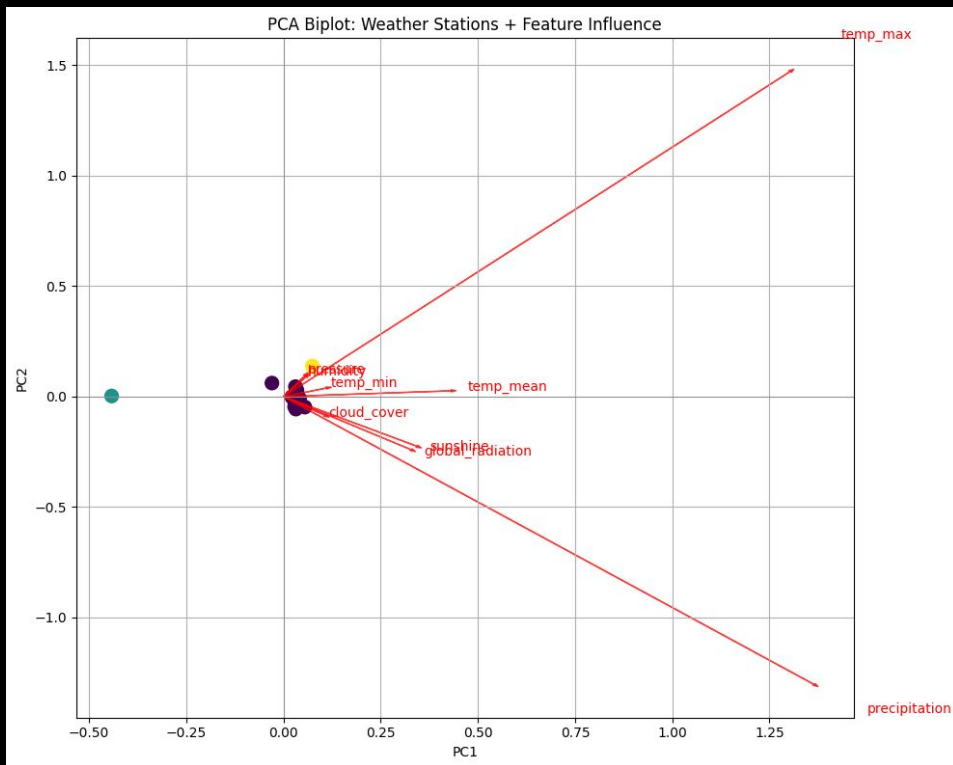
PCA & Clustering of Weather Station Fingerprints



- PCA reduced 9 weather features → 2D space per station
 - Each point = a station's unique "pleasantness fingerprint"
- **Goal:** Identify which stations share similar patterns for predicting pleasant weather
- **Clusters found:**
 - **Sonnblick** and **Valentia** form single-member clusters
 - Due to 0 pleasant days (Sonnblick) or small data (Valentia)
 - **Madrid** stands out in the main cluster
 - Likely due to low importance for **precipitation**
 - Most stations cluster tightly
 - Suggesting shared dominant features: **temp_max** & **precipitation**

PCA Biplot – Weather Metric Influence on Clusters

- This plot overlays weather feature influence vectors onto the 2D PCA projection
 - Vectors show which features “pull” stations in PC space
 - Longer arrows = greater influence on variance



- Key Observations:
 - **PC1** is strongly shaped by **precipitation** and **temp_max**
 - Reinforces earlier findings from heatmap + feature importance
 - **Cloud cover** and **sunshine** show mild, less directional influence
 - Possibly due to interdependence with temperature features
- Interpretation Caution:
 - Shared “pleasantness recipes” across stations may reflect the **rule-based** definition of pleasantness
 - High RF accuracy likely aided by **threshold-learning** aligned with those rules

	PC1	PC2
cloud_cover	0.053101	-0.042395
global_radiation	0.164066	-0.120224
humidity	0.028673	0.044103
precipitation	0.682271	-0.651795
pressure	0.027921	0.047897
sunshine	0.171339	-0.112338
temp_max	0.651590	0.735880
temp_mean	0.215005	0.012928
temp_min	0.054899	0.019143

Recommendations for Climate Tracking & Forecasting

1. **Prioritize precipitation and max temperature**
 - These were the most influential metrics for predicting "pleasant" weather across stations.
2. **Investigate proxy & emergent indicators**
 - Features like sunshine and cloud cover may capture subtler regional variations not encoded in current thresholds.
3. **Zoom in on tightly clustered stations**
 - Vertical spread along PC2 in Cluster 0 suggests other features (e.g., humidity, pressure) might drive second-order differences.